IMPERIAL
SCHOOL OF PUBLIC HEALTH

Investigating the association between clinical depression and medication adherence among working females (aged 18-66) with Type 2 Diabetes in London, England

Group 6 *Research Methods Research Proposal*
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Contents

1 Introduction

2 Methodology

3 Research Plan

4 Discussion & Conclusion

5 References

1 Introduction

Background

Rationale

Aims & Objectives

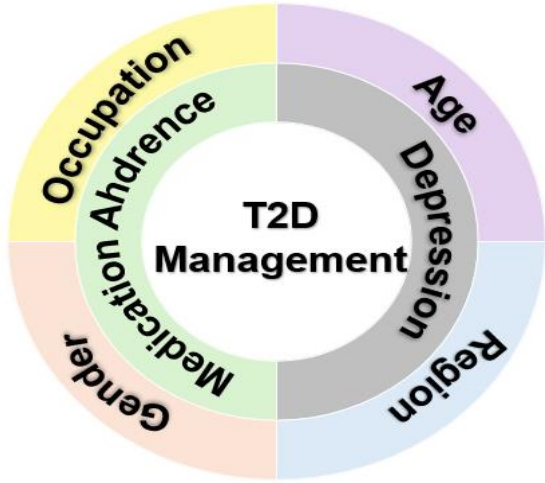
Depression and T2D have a bidirectional relationship, often coexisting

- About 20% of adults with T2D experience depression (1-3).
- Subclinical depression is linked to a 1.7-fold increase in mortality in patients with T2D, but clinical depression was linked to a 2.3-fold increase (4).

Long-term medication adherence is crucial in chronic conditions like diabetes

- Patients often experience significant psychological distress alongside a diminished quality of life, impacting their ability to adhere to medication (5,6).

Factors that affect T2D medication adherence



Population	Working females aged 18-66 in London, England, diagnosed with T2D
Intervention	Presence of clinical depression
Control	Absence of clinical depression
Outcome	Medication adherence to T2D management

Reason for Study

- Depression has been associated with medication nonadherence among females with T2D, which can lead to poor glycaemic control and increased complications (5,8).

Research Gap

- Lack of research on working females (18-66yrs) in London, England.
- Unique challenges faced by women.
- Targeted interventions that consider the social and occupational factors affecting working women.

Research Question: What is the association between clinical depression and medication adherence among working females (18-66 years old) with Type 2 Diabetes in London, England?

Aims

Primary Aim: To assess the relationship between depression and medication adherence among working females with T2D, aged 18-66, in London, England

Secondary Aim: To explore factors affecting medication adherence among working females with T2D

Objectives

1st Objective: To assess the association between depression and medication adherence by comparing working females aged 18-66 with T2D who have depression and those without, over the one-year study period

2nd Objective: To compare the differences in trajectory of medication adherence between working females with T2D who have depression and those without

3rd Objective: To identify barriers to medication adherence among working females with T2D

2 Methodology

Paradigm Perspectives

Study Design

Quantitative Methods

Qualitative Methods

Start
1-year cohort study

Quantitative Methods

Probability based stratified sampling
2 follow-up assessments

Strength of association

Statistical significance

Prospective Cohort Study

Observational Design

6th Months

12th Months

Evaluate

How depression in our targeted group impacts personal adherence to diabetes management

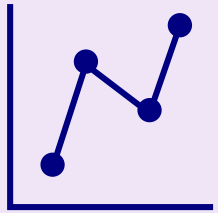
Qualitative Methods

Non-probability-based purposive maximum variation sampling

Yearly assessment

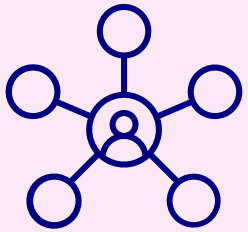
Mixed-Methods
Pragmatism

Quantitative
Positivism



Assesses the association between depression and medication adherence

Qualitative
Relativism



Explores lived experiences of barriers
Influences of cultural and social norms

2.3 Quantitative Methods

Recruitment and Sampling

Data Collection

Data Analysis

Quantitative Methodology: **Compliance Period**

Purpose

To assess participants' willingness and ability to comply with study requirements, such as attending follow-ups or providing complete data.

Duration

2-3 weeks before the formal start of the study

Activities

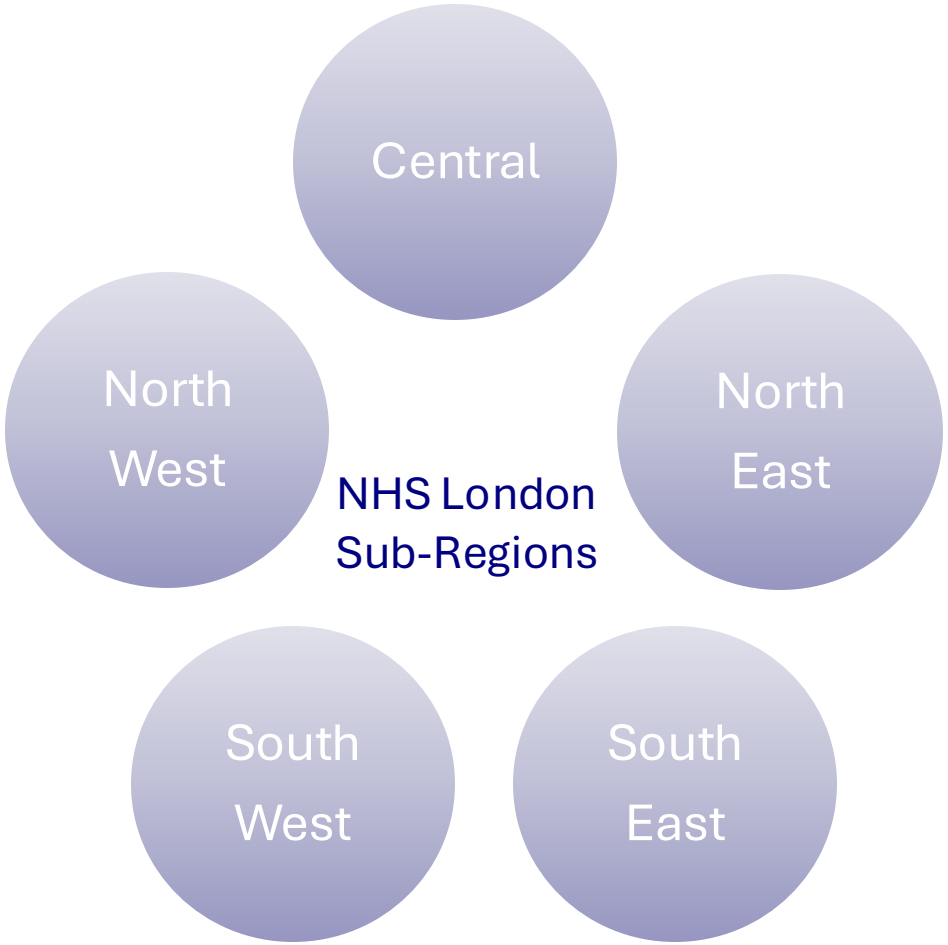
- Complete baseline MMAS-8 surveys for initial adherence.
- Participate in minimal engagement tasks (e.g., attending an orientation, responding to brief questionnaires, or providing initial medical records).
- Consent to preliminary data sharing from their NHS care providers (e.g., blood sugar levels and prescription records).

Handling Non-Compliants and Dropouts

Participants who fail to meet engagement milestones (missing baseline surveys, inconsistent communication) or who formally withdraw will be identified, and collaborated with for insights for qualitative design.

Quantitative Methodology: **Selecting Care Centers**

- **Initial Target Population:** 700 participants to account for pre-study compliance losses and study follow-up attrition
- **Final Cohort Size:** 500 participants for the formal study (250 depressed + 250 non-depressed)
- **Retention Goal**: Proportional representation across sub-regions and depression levels, distributed across NHS GPs and Hospitals



GP Practices

Majority of T2D and mental health management occurs here.

Recruit participants from 2-3 GP practices per sub-region, targeting practices with high patient volumes of T2D and depression to maximize recruitment efficiency.

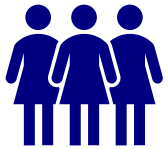
NHS Hospitals

University-affiliated hospitals, like those affiliated with Imperial College London, King's College London, and UCL, provide access to diverse demographics and support streamlined collaboration.

Due to higher patient volumes, 1 hospital per sub-region will be included.

1. **GP Practices:** 10-15 total (2-3 per sub-region; 70 participants per sub-region)
2. **Hospitals:** 5 total (1 per sub-region; 70 participants per sub-region)
3. **Total Recruitment Cites:** 15-20

With 700 participants initially recruited, we aim for 500 participants (100 per sub-region) distributed proportionally across sites to ensure sufficient statistical power while balancing feasibility for follow-up data collection through **probability-based stratified sampling**.



Engagement Strategy:

- Partner with NHS Clinical Commissioning Groups (CCGs) to identify suitable sites and facilitate collaboration.
- Approach GP practices and clinics via regional NHS networks and primary care coordinators to ensure alignment with existing clinical workflows.
- Secure support from university-affiliated hospital research departments and utilize their established patient recruitment systems.



Formal Agreements:

- Formal agreements will outline data-sharing procedures involved, patient identification protocols, and ethical considerations including informed consent.

Significance level (α): p = 0.05 (5%)

- p < 0.05 would mean less than 5% of time the observed effect is not statistically significant and due to chance (Type I error/false positive).

Power (1−β1): β = 0.2; power = 0.80 (80%)

- if there is a true effect, it will be detected 80% of the time and missed 20% of the time.

Confidence level: 95%

Effect size: d = 0.5

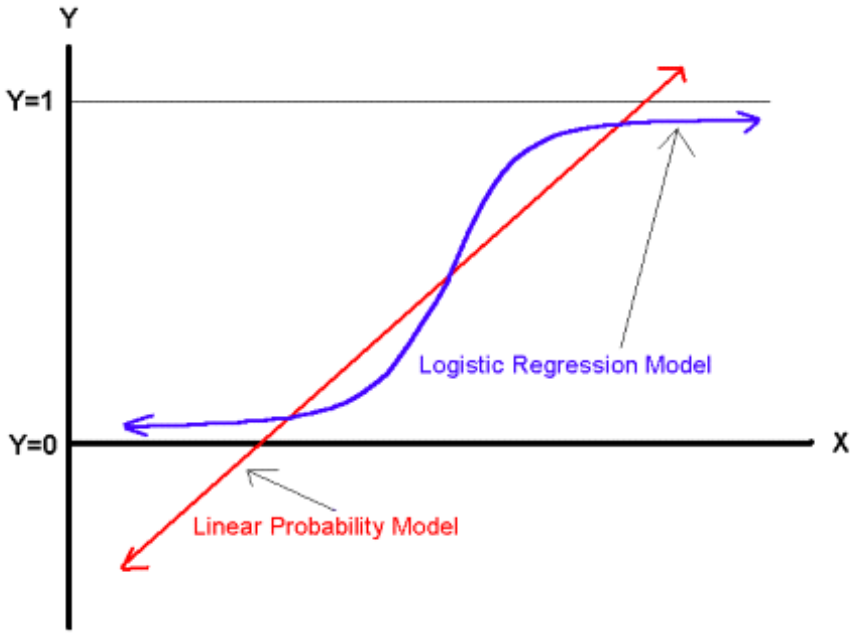
- The expected difference in adherence (measured by MMAS-8) between depressed and non-depressed participant (primary assessment is odds ratio).
- A medium effect size is reasonable because literature suggests that depressive symptoms have a moderate impact on medication adherence.

Inclusion Criteria

- T2D at time of study
- Employed at time of study
- Sex: Female
- Age: 18-66
- Approved consent

Exclusion Criteria

- Non-English speaking
- Other severe psychiatric disorder
- Pregnant
- Recently diagnosed with T2D or clinical depression within 6 months of start of study



Linear and ordinal logistic regression models to estimate relative risk (RR) and odds ratios (OR) for medication adherence

Effect of depression on medication adherence at 12th months

Independent t-test to compare trajectory in medication adherence

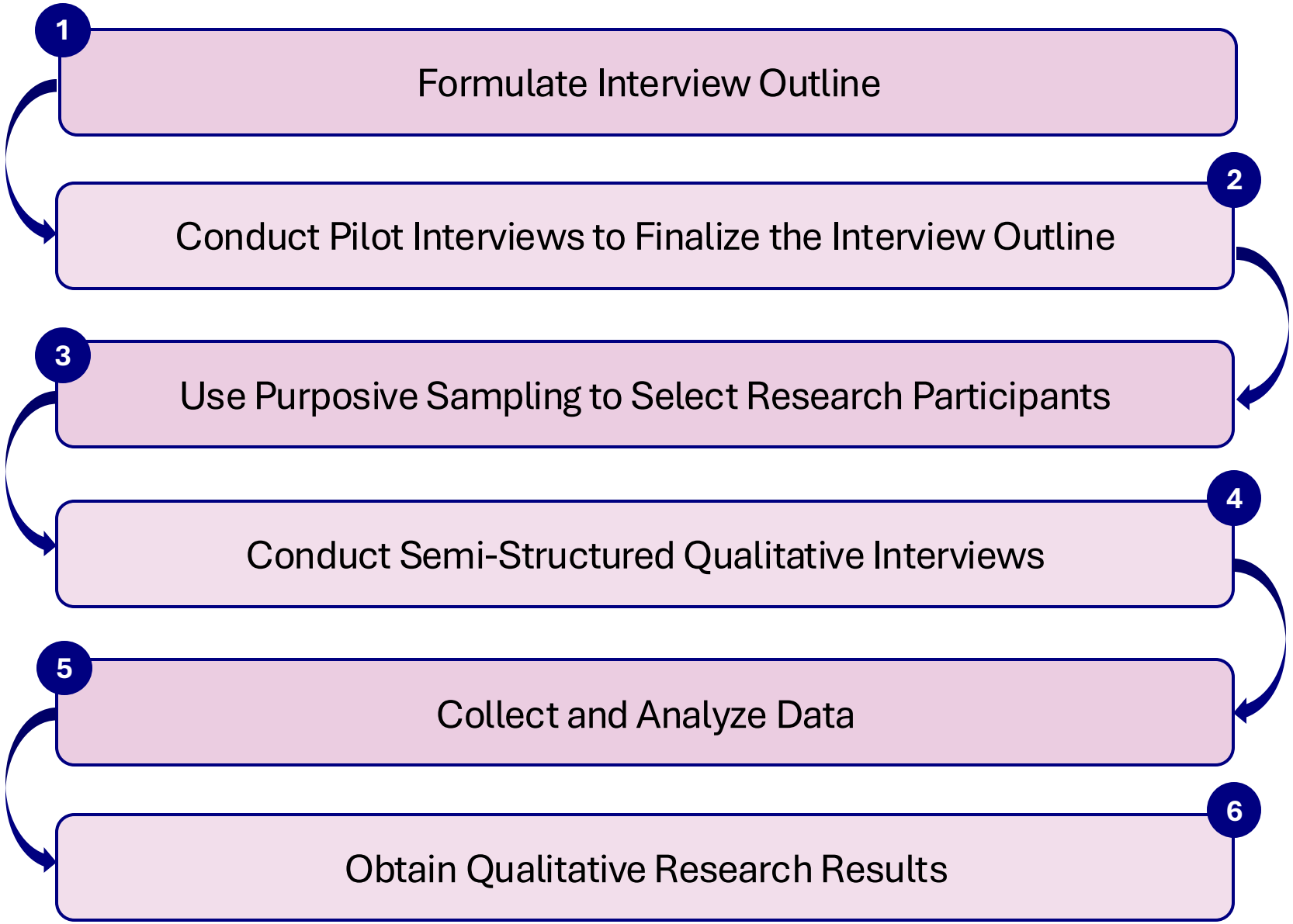
Effect of depression on fluctuations in medication adherence at 6th months and 12th months

2.4 Qualitative Methods

Recruitment & Sampling

Data Collection

Data Analysis


1
Formulate Interview Outline
2
Conduct Pilot Interviews to Finalize the Interview Outline
3
Use Purposive Sampling to Select Research Participants
4
Conduct Semi-Structured Qualitative Interviews
5
Collect and Analyze Data
6
Obtain Qualitative Research Results

Purposive Sampling – Extreme Cases

Intentional selection of participants with low medication adherence to identify barriers among those who struggles most.

Participant Selection

Start with recruitment of 7 participants with depression and 7 without until data saturation is achieved.

Semi-structured Interviews

Use open-ended questions on topics of experiences living with T2D / and with depression, intrinsic and extrinsic factors affecting medication adherence.

01

Deep Understanding

- Personal experiences with depression
- Perceptions and feelings
- Impact on medication adherence

02

Sensitivity

- One-on-one interviews foster open sharing
- Ideal for personal topics like depression and medication adherence

03

Flexibility

- Semi-structured and in-depth interviews
- Explore unexpected themes
- Follow participants' lead for deeper insights

04

Rich, Detailed Data

- Individual stories
- Emotions and behaviors
- Deeper understanding of the issue

Conducting Qualitative Interviews

1

Preparation

- Define a clear research question
- Use open-ended interview guide
- Avoid leading participants

2

Ethics and Consent

- Reconfirm consent
- Ensure participants feel comfortable
- Provide clear information

3

Building Rapport

- Create an inviting environment
- Introduce yourself and the study
- Start with broad, non-invasive questions

4

Flexibility

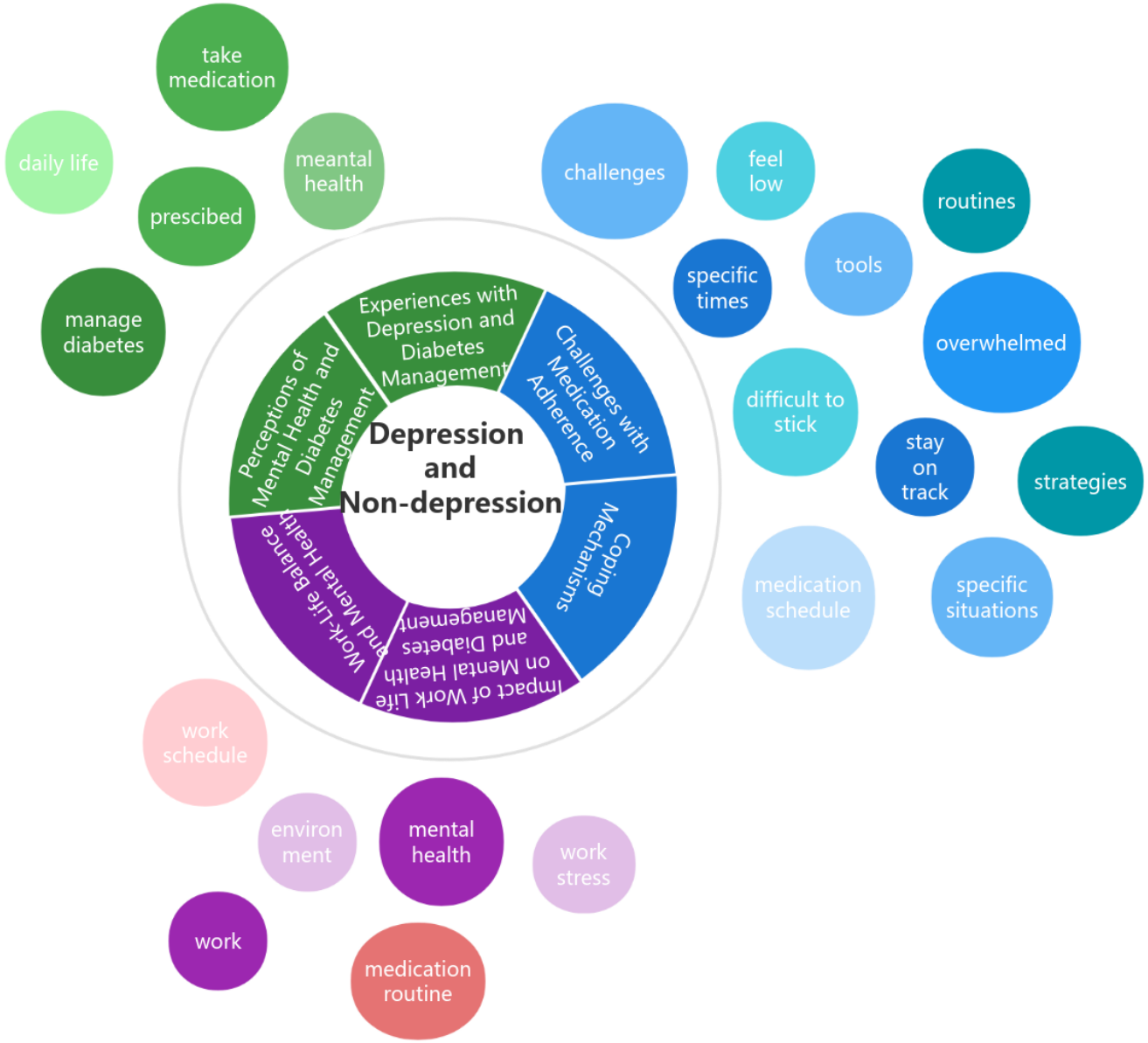
- Adjust questions to responses
- Allow pauses for thoughtful answers

Notes for the Interviewer

- Use **probes** generously.
- Pay attention to **body language** and use active listening techniques.
- Avoid leading or judgmental questions and ensure a neutral tone throughout.
- Respect pauses and give participants time to think.

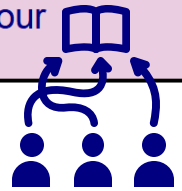
Opening Statement
Hello, my name is _____, and I am conducting research to better understand how clinical depression may affect medication adherence among working women with Type 2 Diabetes. This interview will take approximately 30-45 minutes. I'll ask about your experiences, and there are no right or wrong answers. Everything you share will be confidential. Do you consent to participate in this interview?

Grand Tour Questions	
Non-depression	Depression
Collect participants' demographic information and related disease details. Can you describe a typical day for you, managing your diabetes while balancing work and other responsibilities?	Collect participants' demographic information and related disease details. Could you describe a typical day for you, managing your diabetes while dealing with depression and balancing work?

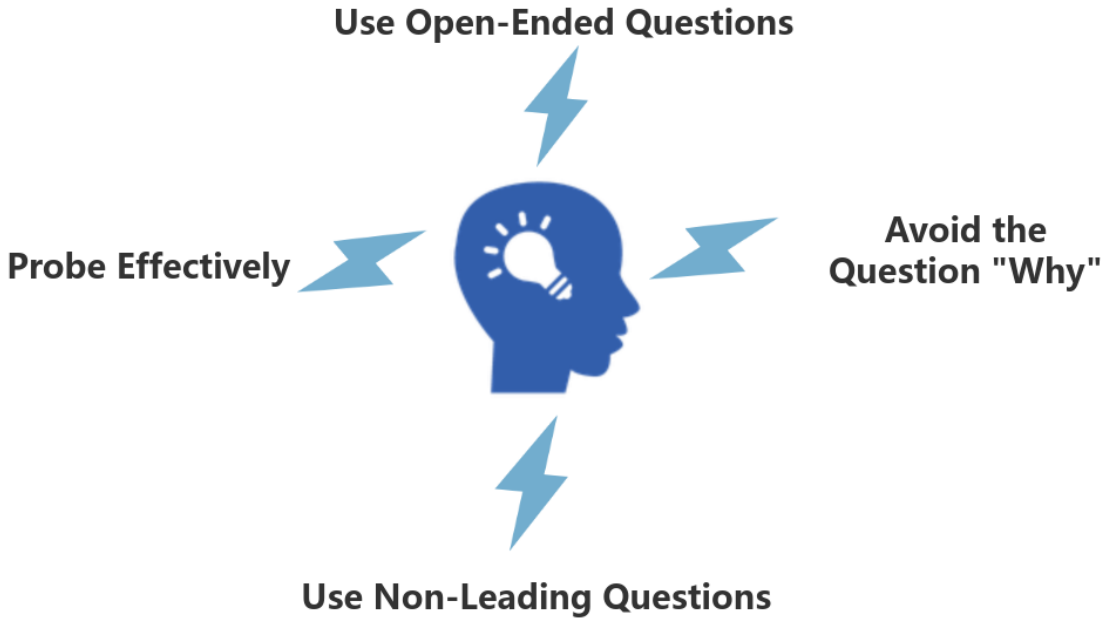


Qualitative Methodology: **Qualitative Data Collection – Example Questions (2)**

Introductory Questions	
Could you share your experiences with taking your diabetes medications? What helps you remember to take your medications, and what challenges make it harder? How do you generally feel about managing your diabetes? Which aspects of managing your condition feel the easiest, and which are the most?	
Non-depression	Depression
Topic 1: Perceptions of Mental Health and Diabetes Management	**Topic 1: Experiences with Depression and Diabetes Management**
Can you tell me about your understanding of the relationship between mental health and managing diabetes? Follow-up: How do you think depression, if present, might influence your diabetes management routine?	How has depression affected your daily life and your ability to manage diabetes? Follow-up: What are some specific ways depression impacts your ability to take medication as prescribed?
Topic 2: Challenges with Medication Adherence	**Topic 2: Coping Mechanisms**
What are some challenges you face when it comes to taking your diabetes medication as prescribed? Follow-up: Are there specific times or situations when it becomes particularly difficult to stick to your medication schedule?	When you feel low or overwhelmed, what strategies do you use to manage your diabetes? Follow-up: Are there specific people, tools, or routines that help you stay on track?
Topic 3: Impact of Work Life on Mental Health and Diabetes Management	**Topic 3: Work-Life Balance and Mental Health**
How does your work schedule or environment affect your ability to manage your diabetes? Follow-up: Do you feel your work influences your mental health, and if so, how?	How does your work environment or schedule influence your mental health and diabetes management? Follow-up: Are there times when work stress has directly impacted your medication routine?



Probing Questions	
Non-depression	Depression
Could you describe a situation where you felt it was particularly hard to follow your medication routine? What factors contributed to this?	Can you describe a time when you felt your depression made it particularly difficult to follow your diabetes treatment plan?
Earlier, you mentioned [specific challenge]. Could you elaborate on what happens during those times? How do you usually handle days when things don't go as planned with your diabetes management?	
Closing the Interview	
Thank you for sharing your experiences. Before we finish, is there anything else you feel is important for me to know about your day-to-day life, or the challenges you face in managing your diabetes and mental health? Your insights have been incredibly helpful. If it's alright with you, I may reach out for a follow-up discussion. Would that be okay? When would be the best time for me to contact you again?	



Collaborative Coding

Interviews coded by 2 independent researchers

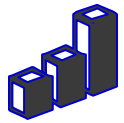
Maximize reliability

Theme Generation

Reflect generative mechanisms and structures influencing medication adherence

Key Features

01  Organize responses systematically

02  Identify recurring patterns

03  Highlight experiences and challenges

Emotional Barriers	Stress, sadness, depressive episodes	→ Skipped doses
Support Systems	Family, friends, healthcare providers' influence	
Work-Life Challenges	Long hours, workload	→ Disrupted routines
Coping Strategies	Reminders, schedules to stay on track	

Review transcripts

Note impressions on mental health & diabetes management

Familiarize with Data

1

Identify key data segments (e.g., "stress impact," "family support")

Generate Initial Codes

2

Group codes into broader themes (e.g., "Emotional Barriers," "Work-Life Balance")

Search for Themes

3

4

Review Themes

Cross-check against data

Ensure clarity & avoid overlap

5

Define & Name Themes

"Emotional Barriers" → Stress/sadness affecting adherence

"Support Systems" → Role of family and healthcare providers

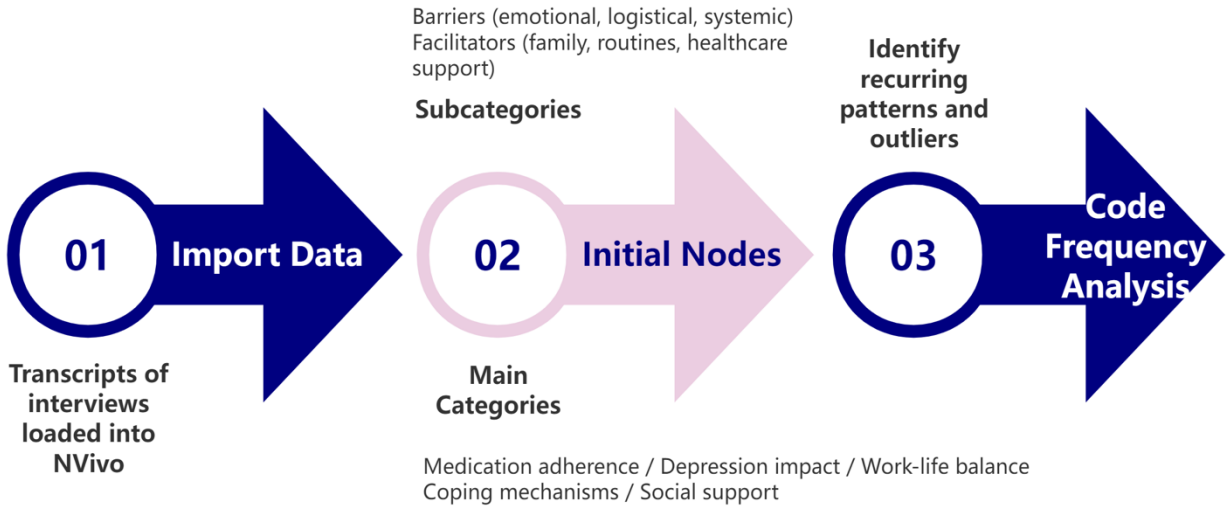
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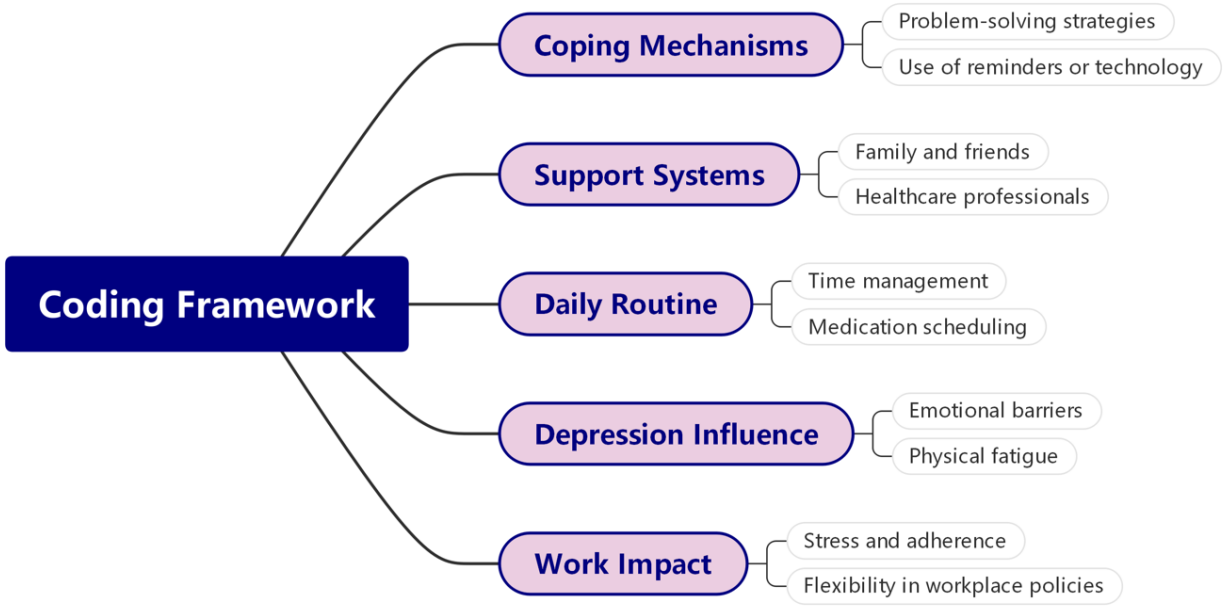
Produce the Report

Link themes to adherence challenges

Use participant quotes for illustration







Key Coding Questions

- How does depression impact medication adherence?
- What strategies help participants tackle challenges?
- How does the workplace facilitate or hinder adherence?
- How do participants view support systems for managing diabetes and depression?

Visualizing Data in NVivo

Coding Stripes

Highlight overlaps between depression and adherence themes

Word Clouds

Visualize common barriers and facilitators



Cross-Tabulation

Compare codes for participants with and without depression

Cross-validation with quantitative findings and other study findings.

3 Research Plan

Ethics

Key Stakeholders

Study Budget

Dissemination Plan

Gantt Chart

Mixed-Methods Design

Informed Consent Form

- Study purpose and procedures.
- Voluntary participation and withdrawal rights.
- Risks and benefits of participation.
- Confidentiality assurance.
- Contact details for questions or concerns.
- Consent statement with signature spaces.

Preparation & Training for Interviewers

- Interviewers will receive specialized training in professional interviewing skills and psychological support techniques.

Support for Interviewees

- A professional psychologist will be available during interview sessions.
- Emergency supplies for participants with diabetes and depression.

Ethics Approval Process

- Since we are funded by the National Institute for Health Research (NIHR), we need to apply through the Integrated Research Application System (IRAS) to the Health Research Authority (HRA) and Research Ethics Committee (REC) to ensure the research complies with ethical and governance requirements.

Access to Medical Records

- **Data Access Request:** Submit a formal research request to the relevant GP practice and the R&D department of the target NHS hospital. Clearly specify the required data (e.g., variables, time range) and its purpose with Informed Consent Form by patients.
- **Compliance and Security:** Ensure all data processing complies with GDPR and NHS data protection policies. This includes:
- Anonymizing sensitive data where appropriate.
- Restricting data access to authorized personnel.
- Using secure systems for data storage and analysis.

Funding Body & Ethics Committee: National Institute for Health and Care Research
Programme: Research for Patient Benefit (NIHR RfPB)

Recruitment: GPs and NHS Hospitals in London

Dissemination

- Diabetes UK
- Diabetes Research & Wellness Foundation
- Advocacy Groups for Women's Health, Mental Health and Diabetes
- Journals: Diabetes Care, BMJ Mental Health

Health Service, Policies & Guidelines

- Healthcare professionals
- Pharmaceutical companies
- Diabetes UK
- National Institute for Health and Care Excellence (NICE)

Research Plan: **Budget**

Item	Cost (£)	Description
Staff		
Psychologists - 2	62,500.00	Entry level psychologist (0.3 FTE) and mid-level psychologist (0.3 FTE), for 3 years
Statistician - 1	24,239.40	0.3 FTE, for 2 years
Research assistants - 2	120,704.00	For 2 years
Interviewers - 4	1,691.20	Based on hourly rates, 2h x 14
Equipment		
General office consumables	615.00	
Laptops - 4	6,000.00	
Recorders - 4	320.00	
Operational		
Advertising leaflets	60.00	
Training costs for interviewers	660.00	
Licensing fee for MMAS-8	1638.00	
Secure cloud data storage	348.30	
Statistical software - R	Free	
Nvivo software	380.00	1 year subscription
Miscellaneous		
Transportation reimbursement	350.00	Compensation for interviewers
Participants' compensation	280.00	
Contingency fund (5%)	10,000.00	
Publication fee	2,450.12	BMJ Open Diabetes Research & Care
Conference fee	293.02	Diabetes Professional Care
Total	232,491.04	

Purpose



- Inform healthcare providers
- Raise awareness for mental health in chronic disease management
- Influence policy-makers to implement mental health screening for diabetes care programs

Target Audience



- Healthcare professionals (diabetes nurses, mental health providers etc.)
- Patient and patient advocacy groups
- Public health authorities focuses on diabetes and mental health programs

Key Messages



- Healthcare providers: Screening for mental health issues can improve patient outcomes
- Patients: Seek help is experiencing depression
- Policy makers: Integrating mental health services into diabetes care programs

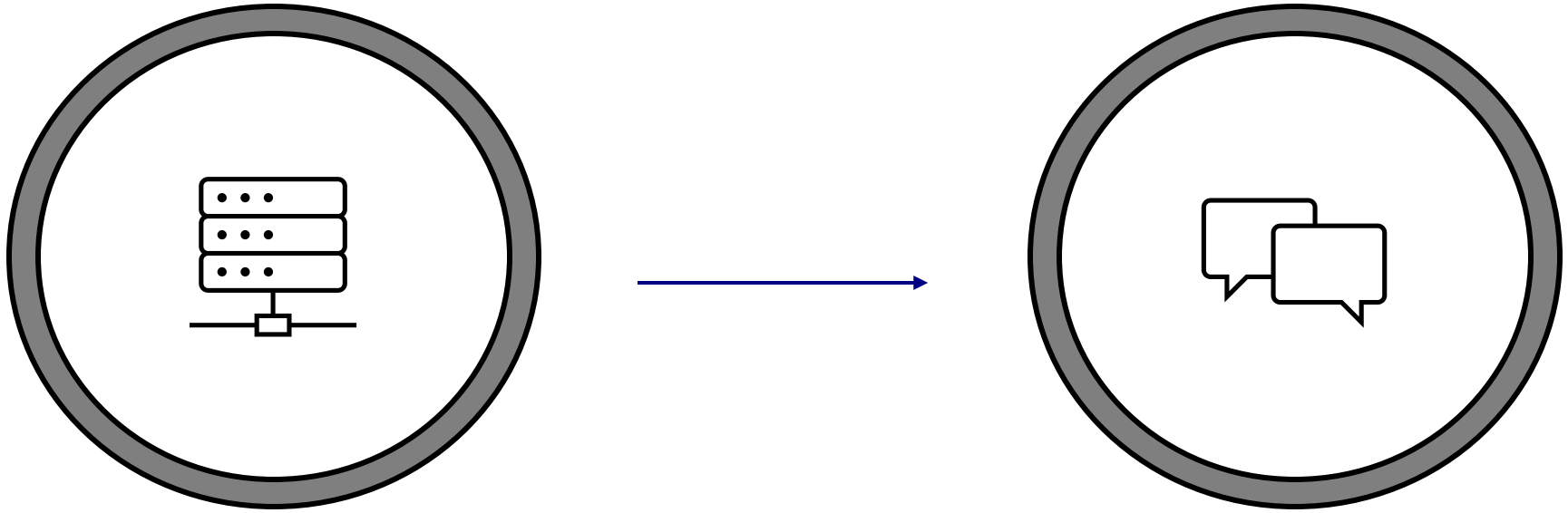
Methods



- Publications: Relevant medical and public health journals
- Conferences: Present research to healthcare professionals
- Health service engagement: Review and improve existing healthcare service
- Community engagement: Mental health organizations can target working women with T2D

Research Plan: **Gantt Chart**

WBS NUMBER	Phases of Study Separated by month	PHASE ONE												PHASE TWO												PHASE THREE				PHASE FOUR								
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
1	**Planning phase**																																				
1.1	Research question																																				
1.3	Secure funding																																				
1.2	Engage with stakeholder																																				
1.1.1	Ethical approval																																				
1.4	Hire researchers																																				
1.5	Compliance period																																				
1.6	Training interviewers																																				
1.7	Material preparation																																				
2	**Quantitative research**																																				
2.1	Data collection																																				
2.2	Data Cleaning																																				
2.3	Data analysis																																				
3	**Qualitative research**																																				
3.1	Data collection																																				
3.2	Data Cleaning																																				
3.2	Data analysis																																				
4	**Completion stage**																																				
4.1	Write up																																				
4.2	Dissemination																																				
4.3	Publication																																				

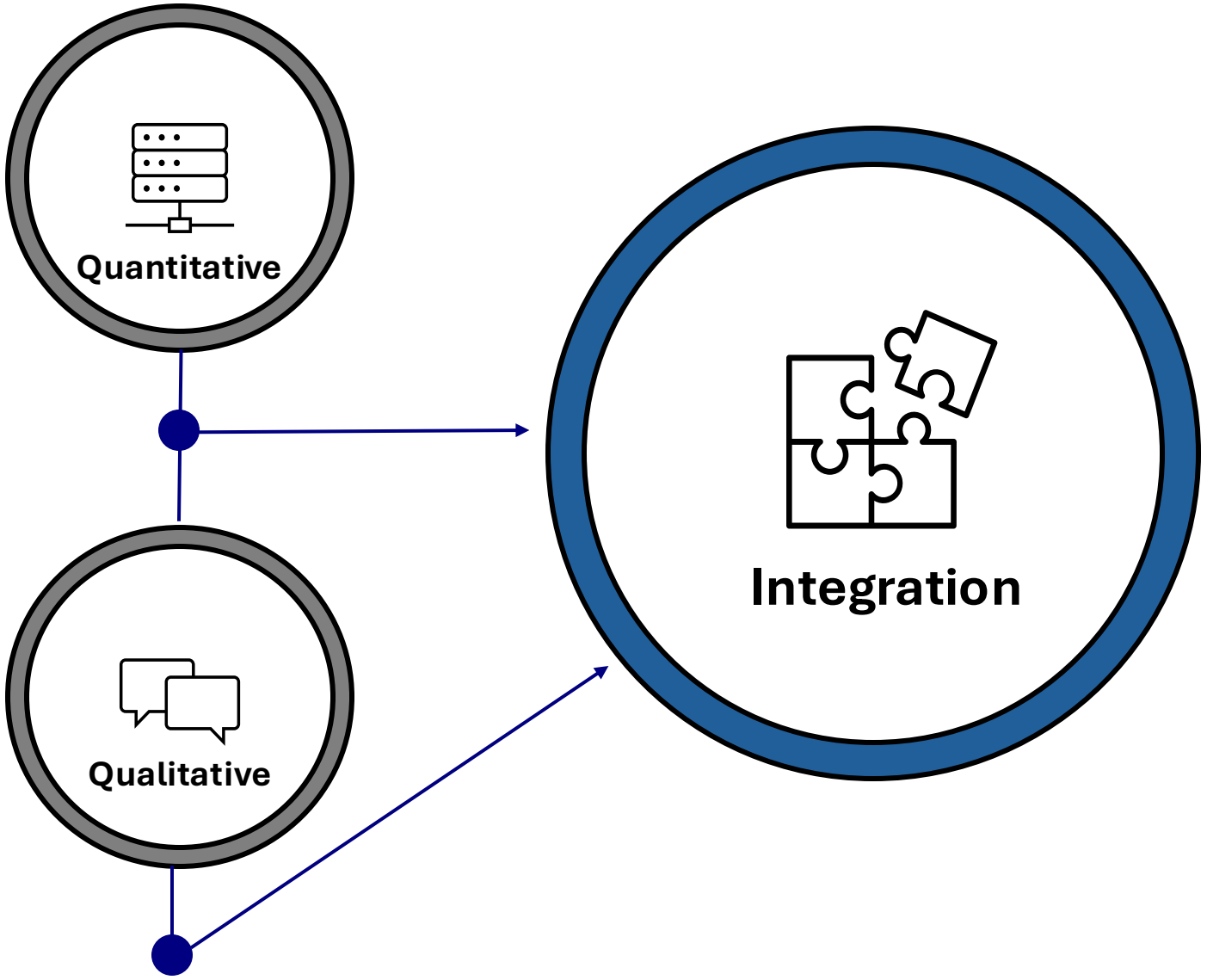


Quantitative

Data analysis to inform selection of interviewees

Qualitative

Semi-structured interviews to explore underlying mechanisms for low medication adherence



- Partially mixed
- Integration at data collection and interpretation
- Contextualizes the observation of low medication adherence behavior

4 Discussion & Conclusion

Key Implications

Strengths and Limitations

Anticipated Challenges

- Provides empirical evidence for medication adherence among working females with T2D and trajectory of changes
- Insights on challenges faced by the target population in diabetes management – social, cultural, gender role

Clinical

- Access to mental health services
- Patient-centred care
- Optimisation of care plans

Policies

- Update on NICE guidelines
- Inclusive workplace policies
- Allocation of healthcare resources

Advocacy

- Awareness of dual burden

Strengths

- Temporal clarity – clear temporal relationship between depression and medication adherence
- Reduced recall bias – data collected in real time
- Strengthened robustness – reduced attrition rate
- Increased generalisability – demographic and socioeconomic diversity

Limitations

- Selection bias – those who are less engaged with the healthcare system and non-English speakers are excluded
- Non-response bias – those with severe depression are less likely to participate
- Measurement error – limitations of the MMAS-8
- Response bias – inaccurate reporting due to social desirability

Stage	Challenges	Anticipates
Recruitment	• Difficulty recruiting working females due to their busy schedules • Participants with depression may feel reluctant to engage	• Use flexible recruitment methods, such as online surveys, evening/weekend interview slots, or workplace outreach • Let trusted doctors or GPs introduce the study, provide materials, explain benefits, and give participants the space to consider
Data Collection	• Self-reported medication adherence may be prone to recall bias or social desirability bias	• Cross-validate with clinical records from GPs or hospitals
Ethics	• Potential high dropout rates due to multiple follow-ups and sensitivity of issues discussed	• Simplify engagement process through flexible data collection methods • Monitor dropout rates and examine causes to address concerns, and improve design and retention of participants • Provide access to psychological support during interviews

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Thank You

Any Questions?

Using Clinical Diagnosis for Depression

- Standardized and consistent criteria which help increase the validity of the measurement.
- Reduces recall bias and subjectivity.

Using MMAS-8 for Medication Adherence

- Has been validated in diverse populations and is widely recognized for its reliability in assessing adherence.
- Addresses various aspects of adherence which provides deeper insight into factors influencing adherence and is useful for intervention planning.
- Brief and straightforward tool for assessment.

Characteristics of Study Participants

- Age
- Education
- Ethnicity
- Employment: full-time / part-time
- Income
- Presence of physical disability or comorbidities
- Years of diagnosis with T2D
- Years of diagnosis with clinical depression
- HbA1c
- Type of T2D medication administration: oral / IV
- Treatment for clinical depression: none / talking therapies / antidepressants
- MMAS-8 score

